# The University of Texas at Austin
# Homework Assignment 11

*Predictive Analytics*

November 13, 2026

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**Instructions:** Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

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## Classification Trees.

**Problem 11.1.** (10 points) Solve Problem **8.4.3** (pp.361-362) in the textbook.

**Problem 11.2.** (10 points) A classification tree is constructed to predict whether a student will pass *Predictive Analytics*. There are two categorical predictors:

- $X_1$ indicating whether the student passed *Linear Algebra* prior to enrolling in *Predictive Analytics*, and
- $X_2$ indicating whether the student passed *Mathematical Statistics* prior to enrolling in *Predictive Analytics*.

Here is the table from a data set of students:

| $X_1$ | $X_2$ | outcome |
|---|---|---|
| 0 | 0 | 5 passed, 20 didn't |
| 1 | 0 | 10 passed, 10 didn't |
| 0 | 1 | 15 passed, 20 didn't |
| 1 | 1 | 15 passed, 5 didn't |

What's the first split made using the Gini index? Be careful to calculate the **weighted** average.

**Problem 11.3.** (30 points) A classification tree is fitted to be used to classify drivers into one of three categories: *Good, Medium* and *Bad*. There are three terminal nodes in the classification tree designating the regions $R_1, R_2$ and $R_3$. Here is the breakdown of categories in each region:

| Region | Good | Medium | Bad |
|---|---|---|---|
| $R_1$ | 70 | 20 | 10 |
| $R_2$ | 30 | 50 | 10 |
| $R_3$ | 20 | 25 | 35 |

Calculate the overall classification error rate, Gini index, and cross-entropy using **weighted** averages.